PRAKTIKUM SISTEM OPERASI

MODUL 10

Shell

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

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JURNAL/TP

1. [40 Poin] Akan dimodifikasi shell dengan modifikasi syscall bernama chname yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall.

   Perhatikan sekarang syscall chname mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

   a. Pada prototypes.h chname diubah menjadi:

```
/* in file chname.c */

extern  pri16   chname(pid32, char *, uint32);
```

   b. Pada chname.c fungsi diubah dari:

```
pri16   chname(
          pid32         pid,            /* ID of process to change      */
          pri16         newprio         /* New priority                 */
        )
```

   menjadi

```
pri16   chname(
          pid32         pid,            /* ID of process to change      */
          char          *newname,
          uint32        namelen
        )
```

   Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil.

   Pengerjaanku ada yang beda dikit, ini kira-kira perubahannya. Jadi ada pemanggilan shell barunya juga, karena kalo ngga, shell baru gabisa dipanggil di minicom.

```
extern shellcmd xsh_namecmd(int nargs, char *args[]);

/* in file chname.c */
extern  pri16   chname(pid32, char *, pri16);
```

```c
#include <xinu.h>
pri16	chname(
	pid32	pid,
	char	*newname,
	pri16	newprio
	)
{
	intmask mask;
	struct procent *prptr;
	pri16	oldprio;

	mask = disable();

	if (pid < 0 || pid >= NPROC) {
		restore(mask);
		return (pri16)SYSERR;
	}
	if (pid <= 0) {
		restore(mask);
		return (pri16)SYSERR;
	}
	prptr = &proctab[pid];
	oldprio = prptr->prprio;

	strncpy(prptr->prname, newname, PNMLEN);

	prptr->prprio = newprio;
```

2. [40 Poin] Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

```c
/* xsh_mycmd.c */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

shellcmd xsh_namecmd(int nargs, char *args[])
{
	if (nargs < 3) {
		printf("Usage: namecmd <pid> <newname> <namelen>\n");
		return 1;
	}

	pid32 pid = atoi(args[1]);
	char *newname = args[2];
	uint32 namelen = atoi(args[3]);

	if (chname(pid, newname, namelen) == SYSERR) {
		printf("Failed to rename process. \n");
	} else {
		printf("process %d renamed to %s (len %d). \n", pid, newname, namelen);
	}
	return 0;
}
```

```c
{"mycmd",		xsh_mycmd},
{"namecmd",		xsh_namecmd},
```

```
xinu@xinu-develop-end:~/xinu/shell$ ls
addargs.c        xsh_cat.c       xsh_help.c      xsh_namecmd.c   xsh_sleep.c
lexan.c          xsh_clear.c     xsh_kill.c      xsh_netinfo.c   xsh_tee.c
Notes_On_Pipes   xsh_date.c      xsh_ls.c        xsh_ns.c        xsh_udpdump.c
shell.c          xsh_devdump.c   xsh_memdump.c   xsh_ping.c      xsh_udpecho.c
xsh_argecho.c    xsh_echo.c      xsh_memstat.c   xsh_ps.c        xsh_udpserver.c
xsh_arp.c        xsh_exit.c      xsh_mycmd.c     xsh_rdstest.c   xsh_uptime.c
```

3. [20 Poin] Test hasilnya:
   a. Masuk ke terminal xinu

```
development-system [Running] - Oracle VM VirtualBox : 1
Activities   Terminal ▾                    Tue 28 Apr, 08:01
                                xinu@xinu-develop-end: ~/xinu/compile
File  Edit  View  Search  Terminal  Help

-----------------------------------------
   __    __   _____   _   _    _    _
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Welcome to Xinu!

xsh $ ?

shell commands are:

argecho      devdump      memdump      ps           udpeserver
arp          echo         memstat      sleep        uptime
cat          help         ns           tee          mycmd
clear        ls           netinfo      udp          namecmd
date         kill         ping         udpecho      ?
```

b. Jalankan perintah ps

```
xsh $ ps
Pid Name              State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ----------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull            ready    0    0 0x005FDFFC 0x00FFFE90      8192
  1 ipout             wait   500    0 0x005FBFFC 0x005FBF30      8192
  2 netin             wait   500    0 0x005F9FFC 0x005F9ED0      8192
  4 Main process      recv    20    3 0x005E7FFC 0x005E7F34     65536
  5 shell             recv    50    4 0x005D7FFC 0x005D79AC      8192
  7 ps_               curr    20    5 0x005F7FFC 0x005F7DE8      8192
```

c. Jalankan perintah namecmd

```
xsh $ namecmd 2 hello 5


New Syscall is easy

process 2 renamed to hello (len 5).
```

d. Jalankan perintah ps

```
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New Syscall is easy

process 2 renamed to hello (len 5).
xsh $ ps
Pid Name              State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ----------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull            ready    0    0 0x005FDFFC 0x00FFFECC      8192
  1 ipout             wait   500    0 0x005FBFFC 0x005FBF30      8192
  2 hello             wait   500    0 0x005F9FFC 0x005F9ED0      8192
  4 Main process      recv    20    3 0x005E7FFC 0x005E7F34     65536
  5 shell             recv    50    4 0x005D7FFC 0x005D79AC      8192
  9 ps                curr    20    5 0x005F7FFC 0x005F7FC4      8192
```

e. Lihat nama proses telah berubah

```
xsh $ namecmd 2 hello 5


New Syscall is easy

process 2 renamed to hello (len 5).
xsh $ ps
Pid Name              State Prio Ppid Stack Base Stack Ptr  Stack Size
--- ----------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull            ready    0    0 0x005FDFFC 0x00FFFECC      8192
  1 ipout             wait   500    0 0x005FBFFC 0x005FBF30      8192
  2 hello             wait   500    0 0x005F9FFC 0x005F9ED0      8192
  4 Main process      recv    20    3 0x005E7FFC 0x005E7F34     65536
  5 shell             recv    50    4 0x005D7FFC 0x005D79AC      8192
  9 ps                curr    20    5 0x005F7FFC 0x005F7FC4      8192
```